

Electromagnetic Method Field Test for Characterizing Steel Fibers in Ultra-High Performance Concrete (UHPC)

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Abstract

The use of ultra-high performance concrete (UHPC) has been growing in the construction industry due to its excellent flexural, tensile, and compressive strengths. The flexural and tensile strength of UHPC is often augmented by the addition of steel fibers. However, the strength of UHPC is highly dependent on the density and orientation distribution of those fibers. Failure to properly distribute the fibers can lead to failures. As a result, steel fiber distributions need to be characterized post-fabrication. Among available testing methods, an electromagnetic technique based on induction sensing had shown significant promise. However, there has been no field experiments to verify fiber density and orientation in real UHPC fabrication. This paper studies the effectiveness of the electromagnetic technique on field-fabricated UHPC. The measurements from our electromagnetic sensor are validated from cores studied with a computed tomography (CT) scan. We demonstrate a 98% correlation between our electromagnetic method and the CT scan results for the orientation, and a root mean square error (RMSE) of 0.295% for the density of the fibers.

Keywords: UHPC, Electromagnet, NDT, Field test

1. Introduction

Ultra-high performance concrete (UHPC) presents opportunities for new types of structural members due to high durability [1], tensile strength [2], and ductility [3]. Often, these strengths are augmented by steel fibers in the mixture. Yet, fibers can segregate due to poor material rheological properties and construction practices [4], leaving zones of high and low fiber content. Fiber orientation can also change during concrete placement, often aligning with the direction of flow. Poorly distributed fibers that are preferentially oriented in an undesirable direction during member fabrication can also lead to structural weaknesses and critical failures [5], [6]. Currently, companies and regulatory bodies have no means to mitigate these hidden weak zones once the concrete member is fabricated unless they know they are present.

In the field of nondestructive evaluation, ultrasound is one of the most common approaches for assessing concrete members [7]. Yet, ultrasound is problematic for assessing fiber content for several reasons. First, there is the problem caused by the internal scattering of the ultrasound because of the interactions with the concrete material. This leads to strong attenuation and dispersive responses from the fibers [7]. The result is that coherent ultrasonic reflections cannot distinguish the steel fibers. Second, even

though the ultrasonic speed in steel fibers is much higher than that obtained in concrete materials, the low percentage of the steel fibers (typically 1% to 3% of the total volume) makes the speed variation, which is the basis of ultrasonic pulse velocity method, not significantly different from that of the bulk speed of the rest of the concrete mixture [8] [9]. Additionally, the variation observed in the speed might also be a result of the larger variations of the aggregates rather than just the steel fibers. As a result, it is difficult to successfully distinguish the steel fibers from the other aggregates. In particular, a feasibility study was published using ultrasound to characterize UHPC, but no variations that could be directly linked to the presence of fibers were detected [10]. Third, conventional ultrasound (which typically has no directional dependence) has no way to estimate steel fiber orientation without some form of tomography system [11] [12], which can be financially and computationally expensive.

Instead, this paper utilizes an electromagnetic approach to steel fiber characterization. Electromagnetic methods have been used in literature to quantify the presence of metals in concrete. For example, eddy current testing has been one of the foremost electromagnetic nondestructive testing methods [13]. In [14], the inclusion of reinforcing steel bars in structures was tested using eddy currents. The eddy current method uses the conductivity of metals to detect the rebar's location. These sensors generate a magnetic field from current that is passed through a coil

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(called the search coil). This induces a current in the steel rebars that creates an opposing magnetic field [15], proportional to the thickness of the steel rebar. However, eddy current has not yet been used in any literature for the quantification of the volume and orientation of steel fibers in steel fiber-reinforced structures. This may be due to the high permeability of the steel fibers, which restricts the eddy current penetration to just the surface of the material under test [13] [16].

Electrical resistivity is another electromagnetic method. This involves placing electrodes on or around the surface of a member to determine how resistant the member is to electrical current [17]. This method had been previously used to determine the volume of steel fibers in steel fiber reinforced concrete (SFRC) [18]. Due to the high conductivity of steel fibers, the resistance of the concrete material reduces proportionally to the volume of steel fibers present in the concrete member [19]. This method is not well suited for field studies because the resistance of the concrete is dependent on physical state, such as the pore structure, the temperature, and the dryness of the concrete [20] [21] [19] [7].

An alternative electromagnetic approach is based on low frequency, inductance-based electromagnetic sensors. These inductive sensors use one or more electromagnets to generate magnetic fields. The magnetic flux of the magnetic field increases in the presence of ferromagnetic materials, which in turn increases the effective inductance of the electromagnet. We then measure this change in inductance. The denser the volume of the steel fibers, the greater the inductance change. Relative to eddy current systems, inductive sensors can usually operate at larger scales and with different materials. For example, eddy current probe ranges are usually < 10 mm and are less effective for ferromagnetic materials, like steel [15]. In the literature, inductance-based sensors have been demonstrated by [22], where an inductive sensor is wrapped around an SFRC member in order to quantify the fiber volume and orientation. In addition, [23, 24] demonstrated a more mobile system similar to ours, where a C-shaped sensor is used to test for the fiber volume and orientation in a lab specimen.

While existing studies have demonstrated the potential of this approach (sometimes referred to as the electromagnetic method) for generic fiber-reinforced concrete, all tests have been in the laboratory with well-controlled specimens [24, 25]. There has been no well-documented study with realistic field specimens. This is important since specimens fabricated in the field can have different and more variable spatial characteristics than those fabricated in the lab. In addition, field specimens tend to be much larger, enabling us to study the effects of concrete flow over larger areas. Due to these factors, a study of sensor performance in field specimens is highly valuable.

This paper provides such a study. We consider fabricated UHPC in five different shapes with different fabrication methodologies. We create a custom electromagnetic

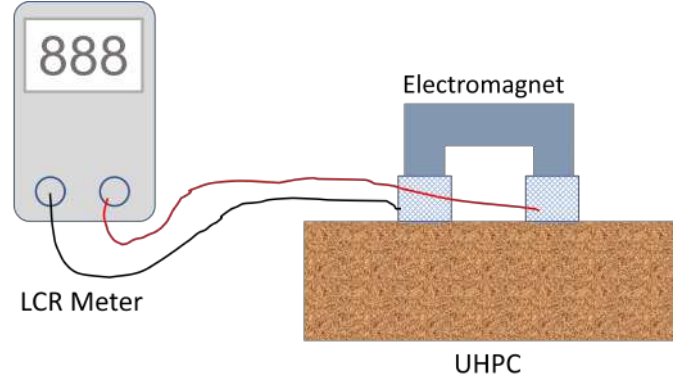


Figure 1: Experimental setup showing the sensor (magnet), the LCR meter which is a BK Precision 880 Dual-Display Handheld LCR Meter, and the UHPC test specimen containing steel fibers

sensor system to quickly measure fiber variations across several meters of a concrete member. We compare these measurements with known features in each of the specimens. We then extract 5 cm cores from regions of interest across the specimens and study the fiber distributions of each core through CT scans. We use these CT scans to validate our sensor readings. The results show that the electromagnetic method performs well and is in good agreement with the CT scan results. Specifically, we achieved a 98% correlation between our electromagnetic method and the CT scan results for the orientation, and a root mean square error (RMSE) of 0.295% for the density of the fibers.

2. Dual Sensor Field System

Our system comprises of two magnets measuring 96 mm in length, 28 by 30 mm in cross-sectional area, and with 210 turns of AWG (American Wire Gauge) 25 each individually connected to an inductance, capacitance, and resistance (LCR) meter. The two magnets are perpendicularly oriented to measure fiber density in two directions. The magnets and LCR meters along with wheels, a rotary encoder, and data acquisition system are all mounted to the enclosure. The enclosure is designed to roll along a surface while keeping the magnets a set height from that surface for sensing. The system directly connects to a computer workstation through a long universal serial bus tether cable. Figure 1 illustrates how this system is configured with a single magnet. Our dual magnets are oriented perpendicularly and are shown in Figure 2.

The magnets are not self-exciting and need an excitation circuit. We use the BK Precision 880 Dual-Display Handheld LCR meter as both the exciter and the reader. The LCR meter sends 50 mA of current through the magnet coils to create a magnetic field. When the magnetic field interacts with the steel fibers, the interaction changes the magnetic field and consequently changes the inductance read by the LCR meter and stored on the computer.



Figure 2: The two magnets used in our sensor setup. The magnet consists of 210 turns of American wire gauge 25 magnetic wire, with an inductance of 7.5 mH

This system also uses an encoder and a data acquisition system. The encoder allows us to collect accurate data relating the position of the magnet with time. The encoder is connected to a wheel that revolves as we move the sensor, and the data acquisition system enables us to collect the encoder data while logging it onto the computer. We use the term inductance change to describe the difference in the inductance relative to the magnets in air. We use this metric to study the distribution of steel fibers.

After the device is turned on, the device is held in air to obtain baseline measurements. The device is then placed on the surface of the specimen that has been cleared of debris, making sure the wheels and the encoder can move as smoothly as possible on the specimen's surface. We then slowly wheel the device along the surface to be scanned while moving the device with the handle, shown in Figure 3. The device collects inductance data and the encoder collects the positional data so we can map the data to position. After this is done, we either take repeated measurements or move on to the next structural member and repeat the same process.

We used two different designs for the enclosure. The first design was built from an existing plastic mold and allowed us to maneuver the sensor system from a long handle and two front wheels. The enclosure was later updated with 3D printed enclosure that is handheld and contains four wheels. This increases the reliability and the durability of the system, since the 3D printed enclosure is sturdier and more durable. The updated version is as shown in Figure 3. It is easier to pick up, drive, and better for scanning vertical structures. In our update, we also increased the number of the coil turns from the previous 210 turns to 400 turns, using the same 25 gauge AWG wire, and the magnet's distance from the specimen was decreased from about 5 mm to about 4 mm.



Figure 3: Second sensor system with two electromagnets, LCR meter, encoder for location measurement, and data acquisition system, all enclosed in a 3D printed case with a wooden handle and wheels

Our data acquisition system consisted of acquisition and data saving loops for the encoder and the LCR meters. We set the number of samples, sampling rate, and buffer size in the software backend to make sure we can collect and store as much information as the devices are able to acquire per second.

3. Experimental Setup

In this section, we describe our experimental setup with which we apply our sensor system. Specifically, we discuss the UHPC members used in our field tests and the CT scanning process used on cores from these members.

3.1. Field Test Experiments

Field test measurements were taken at the Marcus H. Ansley Structures Research Center located at Innovation Park in Tallahassee, Florida. It is a structural research and testing facility for the Florida Department of Transportation (FDOT). Several UHPC structural members were stored in the storage area behind the laboratory and were available for use in the project. This afforded us the opportunity of having a varied set of structures to test and evaluate our methods.

At the test site, we evaluated fiber content and orientation in five (5) different UHPC members. We measured the samples twice. The first test used the initial version of our sensor system and was performed on March 4, 2021. The second test used the updated version of our sensor system and was performed on January 28, 2022. During each visit, we scanned an H-pile, I-beam, square pile, box-beam, and an octagonal pile, which is hollow on the inside.

The tests performed during each visit are detailed in Table 1 and Table 2, respectively. Each member was designed to include approximately 1.5 % to 2 % of fibers by volume.

Table 1: Tests performed during field site visit on March 4, 2021

Shape	Region Scanned	Number of Cores	Thickness [cm]
H-beam	Web	3 times	13
I-beam	Flange	2 times	14
Box Beam	Surface	3 times	11
Octagonal Pile	Surface	2 times	11.7 - 12.6
Bridge Member	Web	0	-

Table 2: Tests performed during field site visit on January 28, 2022

Shape	Region Scanned	Number of Cores	Thickness [cm]
H-beam	Web	2 times	13
I-beam	Flange	4 times	14
Box Beam	Surface	4 times	11
Octagonal Pile	Surface	0	-

For our field tests, we first prepared each member, making sure no debris or loose metals were present. We then took a calibration air measurement four or five times to establish a steady baseline. We then gently scanned along the specimen surface, taking measurements continuously, as illustrated in Figure 5(a). We repeated measurements to study the repeatability of the results. We also took multiple paths to study fibers varying across the surface of the member. After taking these measurements, we marked regions of interest using an oil-based marker. At these locations, 5 cm diameter cores were taken, drilling through the entire material thickness. With the markings, we signified the direction of our scans relative to the specimen. The oil-based marker was used because it had better durability during the coring process. The cores were then analyzed with an x-ray CT system to obtain a measure of fiber density throughout the core. We compared our calibrated magnetic readings to these CT scan results.

3.2. CT Scanning Process

We used x-ray CT to assess the internal structure of the concrete cores. This system penetrates the cores using high voltage x-rays and measures the variations in these x-rays to compute the internal structure as a three-dimensional volume. The three-dimensional volume can then be analyzed to determine the density and orientation of the steel fibers. Note that due to the challenges with fabricating UHPC in the field, the fiber density and orientation will vary as a function of location within the core. In our analysis, we considered averages over a set thickness of 2.54 cm when comparing the fiber density or orientation observed by the CT with the magnetic inductance readings measured by our sensor.

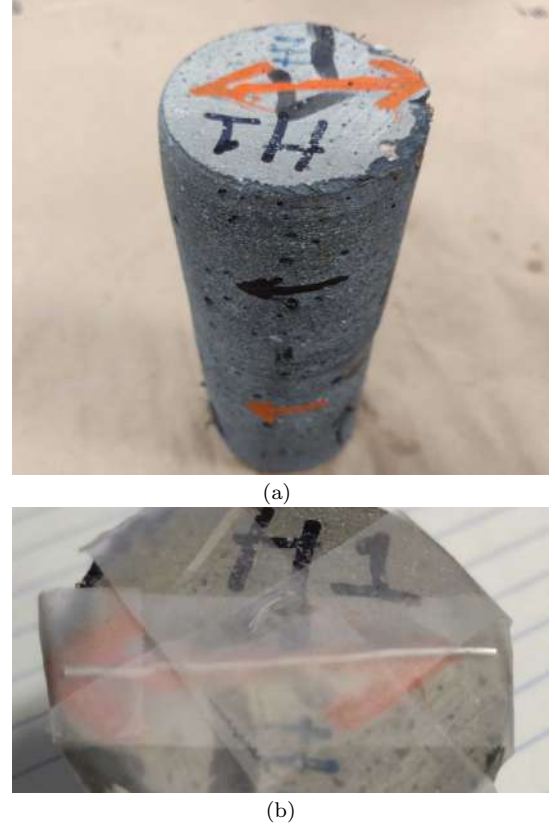


Figure 4: (a) The markings made on the core before taking it out (orange double-headed arrow) and the ones we made right after taking the cores (single-headed black and orange arrows). (b) The aluminum strip held by tapes placed on the double-headed arrow signifying the direction of the magnetic scan.

Here we briefly discuss our process for preparing and scanning each core. Before performing the x-ray CT scan, we placed an aluminum strip on our markings to enable us to define our axes after scanning the material. An example core is shown in Figure 4. Aluminum was used because it is sufficiently dense and can be observed in the CT scan. We then placed this prepared sample into a plastic holder in the CT scanner. We used a plastic holder due to its low density to allow the x-rays to easily penetrate through the holder and not produce any artifacts within the scan. We ensured the sample was stable and did not shift during the scan by adding packing peanuts on the sides. Next we placed the holder with the sample in it into the chuck holder and ensured it sat properly and was tightened securely.

For the scanning, we set our voltage between 200 and 230 kV and our current between 90 and 110 μ A. A higher voltage provides better contrast and penetration depth at the cost of energy consumption and greater x-ray production. Higher currents improve signal-to-noise-ratios (SNR) and can reduce the effect of noise and artifacts. The voltage and current were limited because of machine limitations and because higher voltage and current increases the heating of the x-ray tubes, shortening their life. Higher

voltages and currents than would typically be used to scan concrete were used because of the high cementitious material density and steel fibers. Higher voltages and currents enabled sufficient penetration of the samples with fine resolution. With this configuration, we can have resolutions as fine as $70\text{ }\mu\text{m}$. A high-resolution scan is necessary because the small diameter of the steel fibers ($\approx 200\mu\text{m}$ in diameter) must be distinguishable from the cementitious materials to measure fiber density and orientation.

4. Results and Discussion

In this section, we qualitatively and quantitatively describe results from our field tests. The qualitative results study the measurements taken across each structural member and relate those measurements to known or hypothesized variations. Our quantitative results compare the density and orientation estimates defined by our sensor system and by the x-ray CT system.

4.1. Qualitative Field Test Results

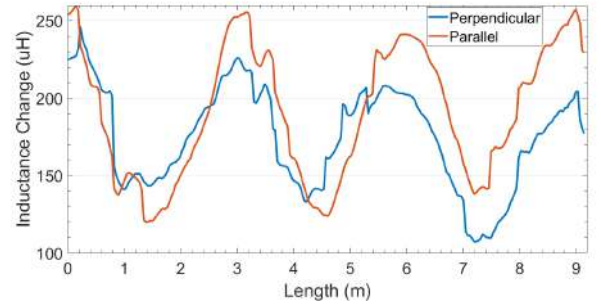
Figure 5(b) illustrates the measurements taken from our H-pile, showing sine-wave-shaped behavior. The two curves describe the measurements from each sensor. The term “parallel” refers to the direction parallel to the member’s long direction (i.e., the direction of travel) whereas “perpendicular” refers to the other sensor. We have found this data to be consistent with repeated measurements. We hypothesize that the sine wave-like pattern is the result of the concrete’s rheological behavior and where the concrete was placed. This results in some portions of the H-pile with a higher inductance change at the peaks of the sine wave due to a higher concentration of the steel fibers there and portions with a lower inductance change at the troughs of the sine wave due to a lower concentration of steel fibers, possibly caused by fiber segregation.

Figure 6(a) shows a scan across the top of the square pile and Figure 6(b) shows the measurements from a scan across the side of the square pile. When the pile was made, it was cast with a form joint, which has visible steel fibers protruding from the surface, as shown in Figure 6(c). There is a visually confirmed higher concentration of steel fibers in this location, which our results confirm. That is, in Figure 6(a) and Figure 6(c), we observe a spike in the inductance change at the location of the joint. Note that there is a difference in the locations of the form joints in the side compared to the top of the square pile, as can also be seen in the results. We also saw a higher concentration of the steel fibers in one of our two sensors, indicating preferential alignment parallel to the length of the square pile.

The scanned octagonal pile had a surface texture on a portion due to a form liner used during placement, as shown in Figure 7(a). Figure 7(b) shows our measurements of this octagonal pile. The region that contained the form liner showed a significant difference in the readings. The



(a)



(b)

Figure 5: (a) The field setup with the device on the H-pile. (b) The result from the two magnets, the parallel and the perpendicular scans are shown. We can see the sine-wave like results, showing high inductance and low inductance points in the H-pile corresponding to high and low relative fiber concentrations.

region without the form pattern had a significantly higher inductance change when compared to the region with the form pattern. We believe the form pattern likely pushed the steel fibers lower into the concrete. As a result, the fibers have a weaker effect on the magnetic readings in regions with the form pattern.

We also tested a bridge member that contained self-compacting concrete (SCC) with no steel fibers in its center and steel fiber-reinforced UHPC at the ends. This is illustrated in Figure 8(a). Our scans indicate that we can distinguish the two regions: the SCC without fibers and the UHPC with fibers. Figure 8(b) shows that the inductance change in the regions with SCC was close to zero, confirming the absence of steel fibers in the SCC regions when compared with the UHPC regions.

4.2. Quantitative Field Test Results

To quantify the volume and orientation of fiber for validation, cores were taken as part of the field tests. From each core, we extracted the CT data, preprocessed the data using the Phoenix Datos-X software, and then exported the data to a 3D file to be analyzed using VGStudio Max [26]. We performed the fiber analysis in VGStudio Max

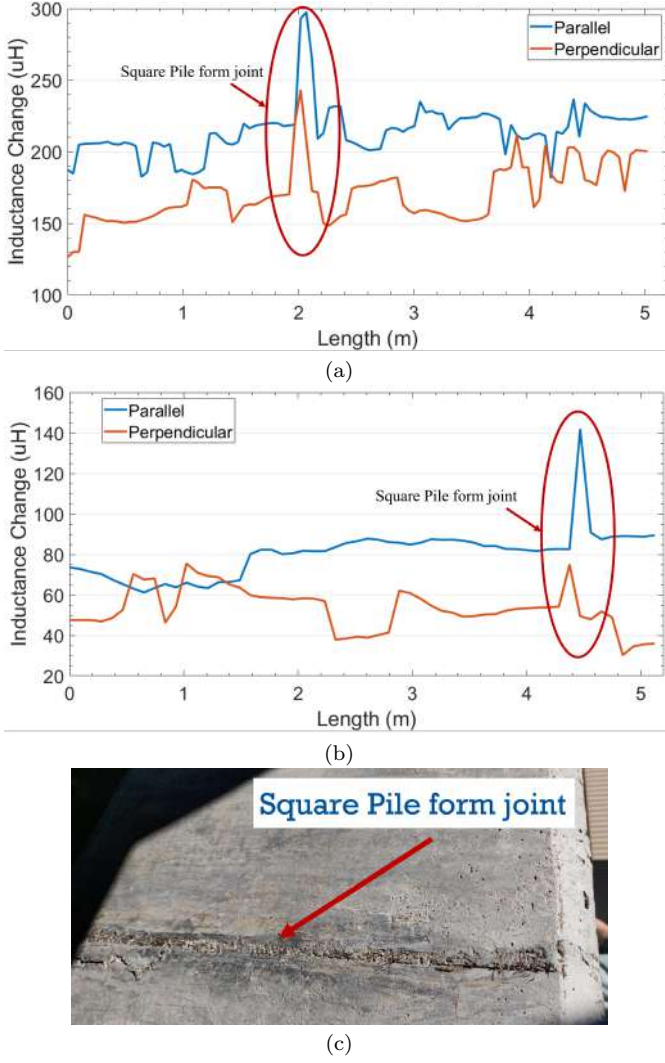


Figure 6: (a) Inductance change for the square pile with a concentration of steel fibers along a form joint on the top of the pile. (b) Inductance change for the square pile with a concentration of steel fibers along a form joint on the side of the pile. (c) Photograph of concrete at location of form joint.

to determine the fiber percentage and the fiber alignment of the top 2.54 cm thickness of the core. We set the voxel grayscale threshold for the fibers so that every voxel with a grayscale value above this threshold was considered to be part of a steel fiber. The threshold was determined from a fiber thickness analysis in which we use the ray method of wall thickness analysis in VGStudio Max. This fiber thickness analysis provides the average fiber thickness across the core. Figure 9(a) shows an example of the fiber distribution thickness for a core from H-pile. This gives an idea of how accurate our thresholding using the fiber thickness method.

We determined the threshold value by measuring steel fiber diameters with an optical microscope to digitally measure and record the thickness of 50 individual steel fibers to ensure that we have the correct fiber thickness and hence, that we set a good threshold for distinguishing

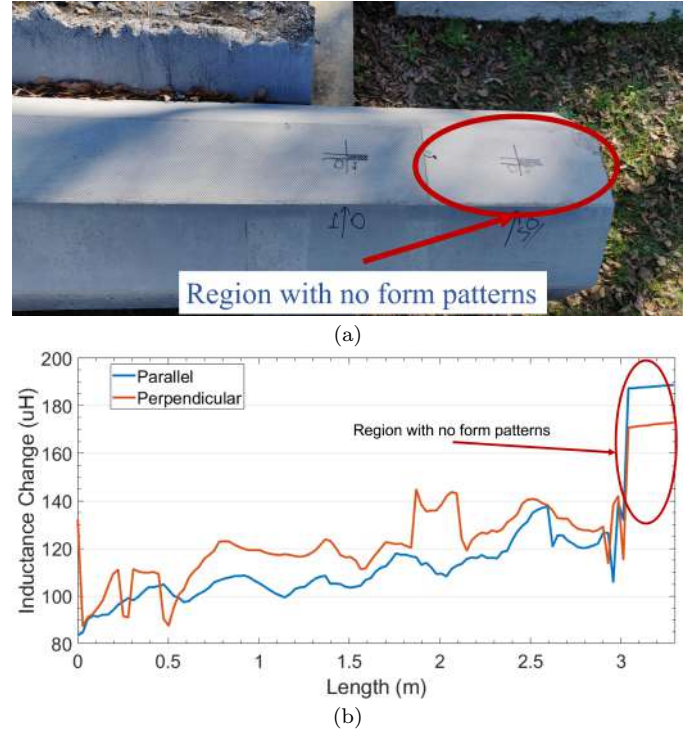


Figure 7: (a) Octagonal pile highlighting region without surface texture imparted by form line. (b) The form pattern on the surface of an octagonal pile and the corresponding inductance changes. The regions with the form pattern had a lower relative inductance change (steel fibers) due to the perceived pushing below the surface by the form pattern structure.

the steel fibers from the cementitious materials. The fiber thickness histogram is shown in Figure 9(b). A reference image of such a typical steel fiber is shown in Figure 9(c), showing both the scale and measured thickness. The average thickness was $210 \mu\text{m}$ and the standard deviation was $3.14 \mu\text{m}$.

Based on the fiber distribution measured in the microscope, we chose a threshold value that provided an average fiber diameter of $200 \mu\text{m}$. This threshold value was then used in our remaining volumetric analysis to determine the fiber percentage and fiber orientation. The fiber percentage was obtained by calculating the number of voxels that were steel fibers as a percentage of the total number of voxels that comprised the whole UHPC member. The orientation was calculated by determining the number of voxels of steel fibers that were oriented along predefined axes, defined by the aluminum strip on top to the core, which is parallel to the member's length. These numbers were plotted against the angular displacement of these voxels from the respective axes. For example, from an x-axis reference point, all the fibers along the 0° axis are perfectly parallel to the x-axis and all the fibers shown at 90° were perfectly perpendicular to the x-axis.

Figure 10 shows the orientation of the fibers in each of the axes. Scans with more blue fibers show more fibers in the direction of the respective axis. From these scans, we

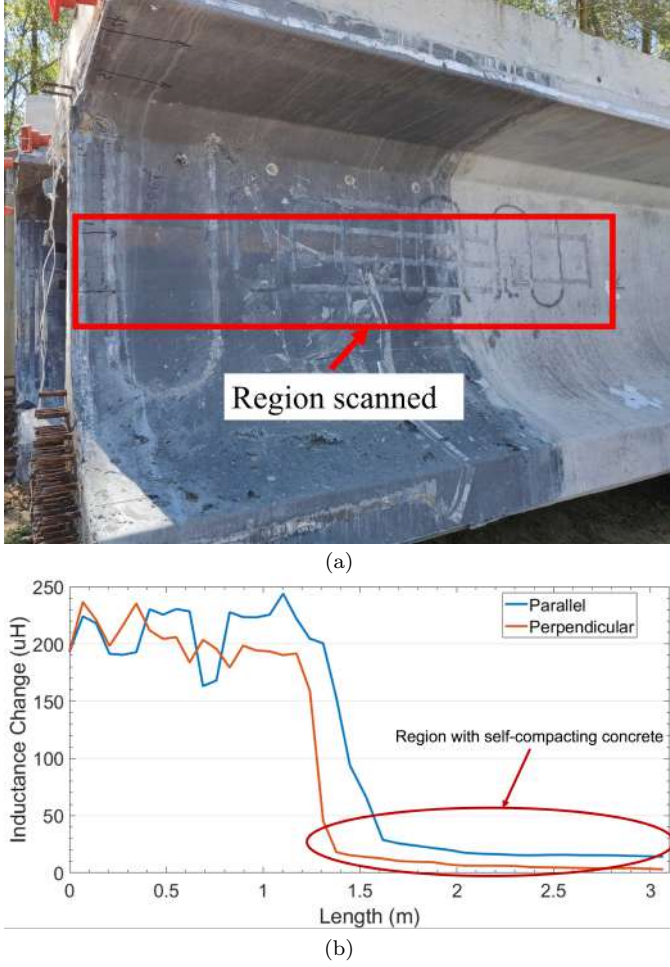


Figure 8: (a) The bridge member G6E1 with the darker portions (end portion) as the UHPC and the lighter portion (remaining portion) as the SCC without fibers. (b) The result of the scan of the first portion of the bridge member showing the clearly delineated UHPC - SCC regions clearly marked by the inductance change going to zero or slightly below zero.

see that there were distinct layers of concrete with preferentially oriented fibers. To measure fiber orientation, we calculated the ratio of the values for the x-axis to the values for the y-axis to measure the relative orientation. If the value is greater than 1, the fibers are more aligned towards the x-axis. If the value is less than 1, they are more aligned towards the y-axis. We apply a similar approach to our magnetic inductive sensor data.

4.3. CT Scan and Magnetic Sensor Comparison Results

For the magnetic fiber percentage, we calibrated the fiber percentage to the inductance change using lab specimens. This was done by measuring the inductance change of multiple samples with known fiber percentages and finding the linear relationship between them. The equation that describes their relationship in the form $y = mx + c$ is then used as the conversion factor for the field test results, where y is the field test percentage, x would be the inductance change, and m is the slope of the line that

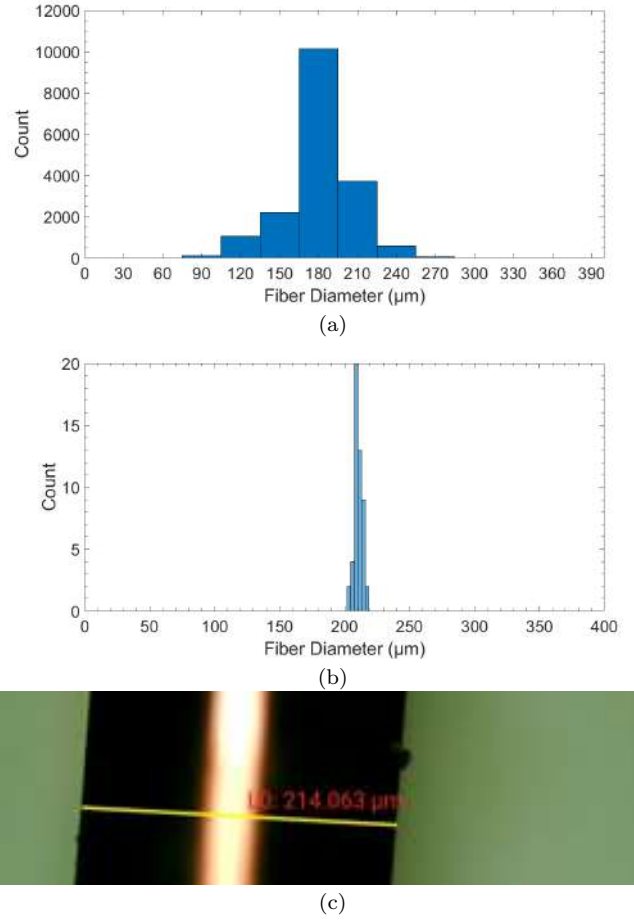


Figure 9: (a) Fiber thickness distribution in a core, based on the threshold chosen in VGStudio. The mean thickness of the fibers is 200 μm is in line with the expected thickness. (b) A histogram of the steel fiber diameters measured and plotted, giving the average thickness as 210 μm. (c) A figure showing the measured diameter of a typical steel fiber used.

describes the relationship between the inductance change and fiber percent of the lab specimen. We compute this separately for each sensor and then compute the average as our fiber percentage.

From this calibration, we obtain the magnetic percentage for our field result, and from these we compared the results of our magnetic scan and the CT core scans. Similar to our calculation of orientation from the CT scans, the magnetic fiber orientations were obtained by calculating the ratios of the perpendicular magnetic reading to that of the parallel magnetic reading. This serves as the relative orientation, with values greater than 1 meaning that the relative orientation of the fibers are more perpendicular than parallel, and vice versa.

Figure 11 compares the relative orientation values from the magnetic measurements and the CT scan. The results show a good agreement with an R-squared value of 0.978. The best linear fit line nearly passes through (1,1), where both directions experience the same inductance change (i.e., there is likely no preferred orientation).

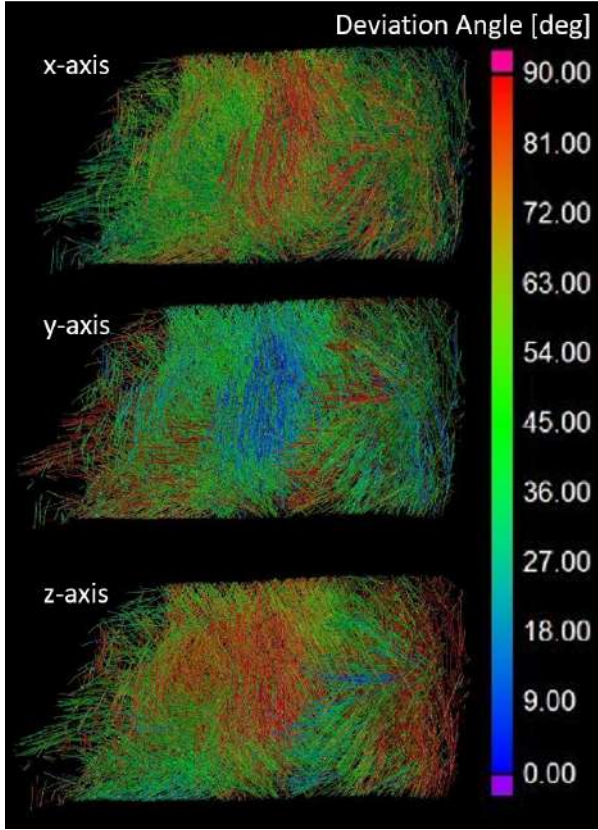


Figure 10: The fiber orientation analysis with CT Scanner. Fiber orientation was analyzed in the three directions: x , y, and z. For each of the axes, blue means the fibers are aligned with the axis, red means that the fibers are roughly perpendicular with the axis, and green means that the fibers are in intermediate range.

However, there is a deviation in the slope from 1. This could potentially be improved in the future by calibrating our magnetic readings with CT scans of known fiber percentages.

Figure 12 compares the fiber percentages computed by the magnetic sensor and the CT scan. We observe a linear relationship between the fiber percentages for the CT scan and the magnetic sensor readings with an R-squared value of 0.207. Despite the noisy relationship, our measured magnetic results have an overall 0.295% root mean squared error (RMSE) with our CT measurements. Based on existing literature [27, 28], estimation errors below 0.5% are generally considered sufficiently close estimates in fiber percentage. The spread on the data is larger than observed with the orientation. We believe this is because the orientation of the fibers, especially the fibers perpendicular to the concrete surface, will affect the magnetic field measured and consequently the fiber percentage calculated. As a result, more variability due to the inhomogeneity of the concrete is expected. Overall, both of our quantitative results show that the electromagnetic method is effective at estimating fiber orientation and percentage.

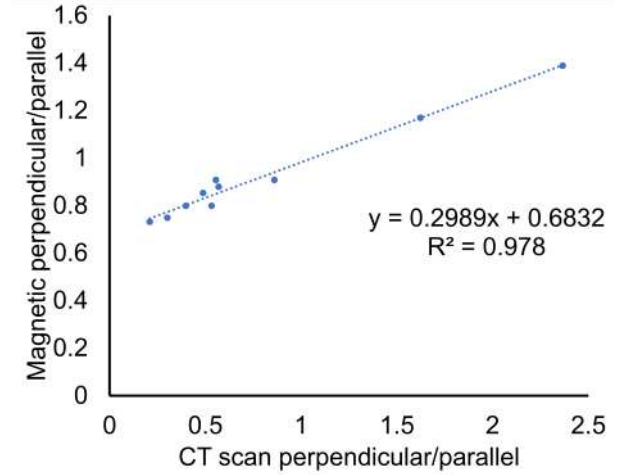


Figure 11: The comparison between the magnetic orientation result versus the CT scan orientation result. The perpendicular/parallel is a ratio identifying the relative orientation of the steel fibers.

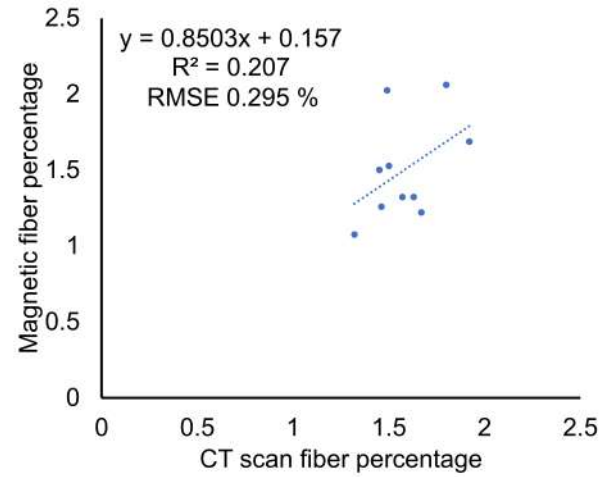


Figure 12: The comparison between our magnetic fiber percentages of the cores taken versus the fiber percentage obtained from the CT scans.

5. Conclusion

In this work, we tested an electromagnetic sensor based on inductive principles to characterize large ultra-high performance concrete (UHPC) members fabricated in the field to quantify the steel fiber percentages and orientation. Our results demonstrated excellent capability to distinguish medium to severe variations in fiber percentage and orientation. This was demonstrated both through qualitative observations with the system as well and quantitative analyses, comparing our magnetic readings with values extracted from CT scans of cores from our field tests.

For future work, we will improve the portability of our design for broad field tests and use by technicians out on the field. In addition, future optimizations can be made to the sensor design (e.g., switching to a soft iron core) and circuitry/analysis algorithms (e.g., leveraging multiple frequency information) to allow us to achieve more reliable

measurements.

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